hedged_blend_s10

portfolio strategy | family: portfolio | status: research | derived from `hedged_blend`

What this strategy does

The same blend with the stronger research equity sleeve (s10_vol_rank_tilt: Sharpe 1.8)

Why it should work. As hedged_blend, with the volatility-rank equity book as the alpha sleeve.

Design notes (from the strategy's source file).

PortfolioStrategy: a fixed-weight, daily-rebalanced blend of other registered strategies.

A sleeve weight is a multiple of fund NAV allocated to that strategy's own sizing: weight 1.0 runs the strategy exactly as backtested on the whole NAV, 0.8 runs it 20% smaller. Weights need not sum to one -- the sleeves are low-correlation books that each use only part of their capital (the options sleeve spends at most its premium budget, the equity sleeve is margin-financed long/short), so they can share one NAV. Returns compound daily: r_p = sum(w_i r_i), NAV_t = NAV_{t-1} (1 + r_p).

Sleeve results come from the strategy registry itself (re-run in-process, ~seconds), so the blend always reflects the current sleeve definitions. Days with no sleeve activity (options in 2011, where the source file is corrupt) contribute zero for that sleeve.

parameter	value
w_s10_vol_rank_tilt	0.9
w_opt_kelly_cap20	1.6

Performance

metric	hedged blend s10	market (CRSP VW)
Annualized return	12.43%	9.94%
Annualized volatility	4.73%	19.35%
Sharpe (NAV-pct)	2.50	0.59
Sortino	4.08	
Max drawdown	-15.5%	-55.5%
Calmar	0.80	
Final NAV ($10M start)	$97,913,566	
Worst day	-2.04%	
% positive days	54.2%	
Sharpe 1996-2001	2.92	
Sharpe 2002-2007	2.84	
Sharpe 2008-2013	2.10	

Correlation of daily returns with the market: **0.12**.

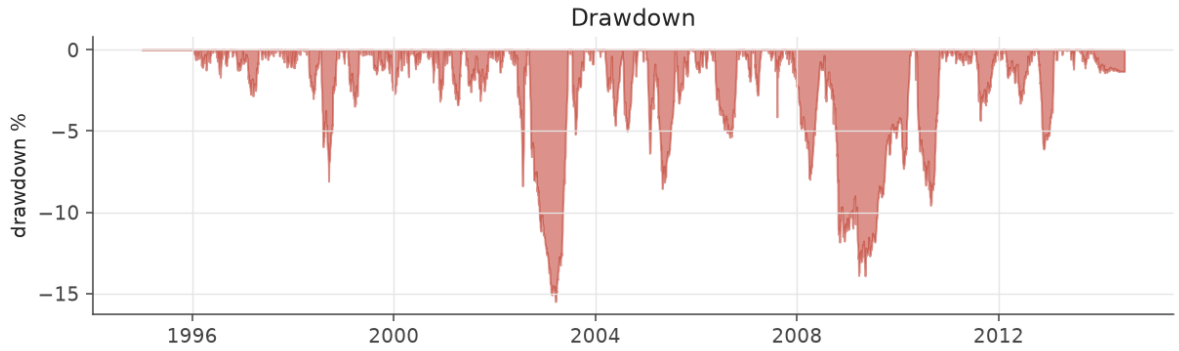
• best Sharpe among portfolio: this strategy; this one: 2.5

• best return among portfolio: this strategy; this one: 0.124

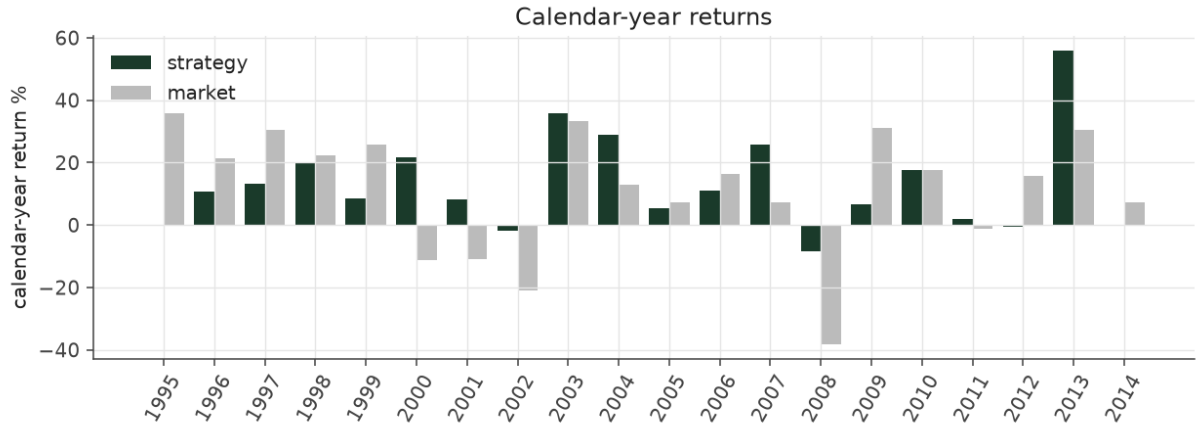
• best Calmar among portfolio: this strategy; this one: 0.802



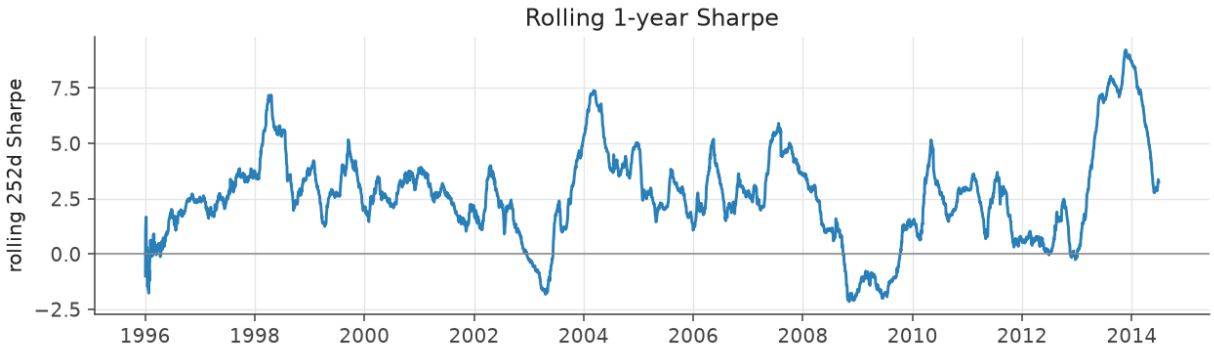
NAV (log scale)



Drawdown from running peak

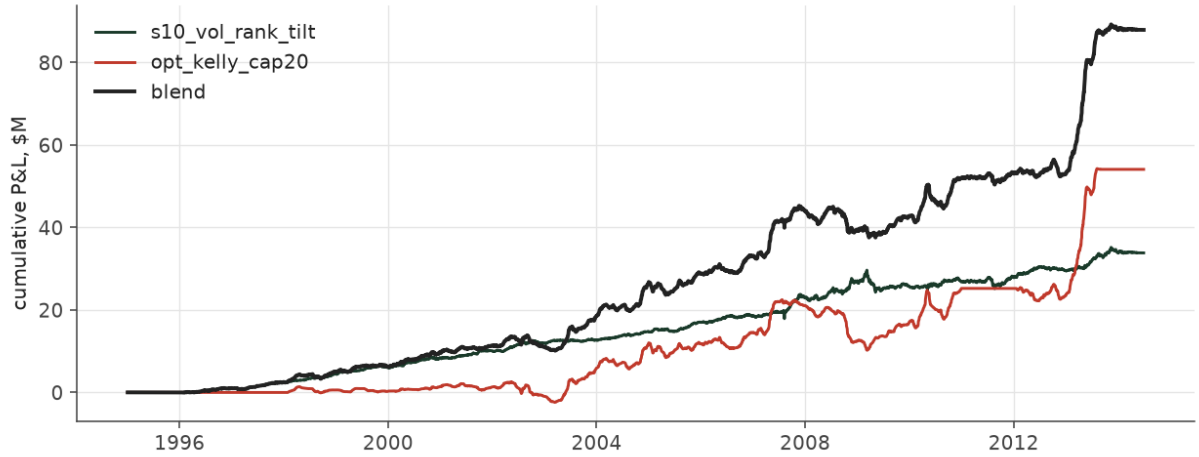


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Each sleeve's contribution to the blend's cumulative P&L

sleeve	weight	pnl M	share of pnl
s10_vol_rank_tilt	0.90	33.83	38.5%
opt_kelly_cap20	1.60	54.09	61.5%

• Correlation between the sleeves' daily returns: **-0.05**.

• Blend market beta 0.030, correlation to the market 0.12.

Return, volatility and Sharpe of the blend are in the Performance table above; the cross-strategy report (`python -m pead compare`) shows the full weight frontier, the walk-forward (out-of-sample) weights and the volatility-target table.